

Question 1: (Start a new Booklet)

- (i) Given that: $I = \frac{3E}{R+r}$, and $E = 1.5$, $R = 1.5$, $r = 2.5$, find I .
- (ii) For the points A (2,-3), B(-3,1) find:
 (a) the gradient of the line AB
 (b) the angle of inclination of the line AB, to the positive direction of the x axis.
 (Answer to nearest degree)
- (iii) Simplify: $\frac{x^2 + x - 12}{x^2 - 9}$
- (iv) Show that: $\frac{1}{3-\sqrt{2}} + \frac{1}{3+\sqrt{2}}$ is a rational number.
- (v) Solve: $|3x - 7| = 8$
- (vi) Write $0.04\dot{1}$ in simplest fractional form.

Question 2: (Start a new booklet)

- (i) The quadratic equation $3x^2 + 4x - 3 = 0$ has roots α and β .
 Without finding α and β , evaluate the following:
 (a) $\alpha + \beta$
 (b) $\alpha\beta$
 (c) $\alpha^2 + \beta^2$
- (ii) Write in simplest surd form,
 (a) $\frac{\sin 60^\circ}{\cos 60^\circ}$
 (b) $\cos 225^\circ$
 (c) $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$
- (iii) Prove that the line $4x + 3y + 18 = 0$ is a tangent to the circle, centre (-1,2) and radius 4 units.

Question 3: (Start a new booklet)

- (i) Solve : $4 \tan \theta = -3$, for $0^\circ \leq \theta \leq 360^\circ$
- (ii) For what values of k will the equation $x^2 - (k+4)x + (7+k) = 0$ have equal roots.
- (iii) Find the indefinite integral of:
- (a) $\frac{1}{x^3}$
- (b) $(5x-2)^3$
- (iv) Differentiate with respect to x ,
- (a) $y = \frac{(x+5)}{(1+3x)}$
- (b) $y = \log_e(4-3x)$

Question 4: (Start a new booklet)

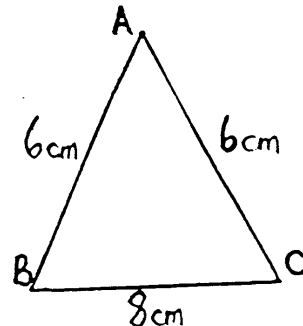
- (i) Sketch: $y = |x-3|$, showing all main features.
- (ii) Solve: $\frac{9-Y}{3} > 2 - \frac{Y}{5}$

- (iii) In the diagram (not drawn to scale),

$$AB = 6 \text{ cm}$$

$$AC = 6 \text{ cm}$$

$$BC = 8 \text{ cm}$$



- (a) Find $\angle CAB$ to the nearest minute.
- (b) Hence, if D is on AC , such that $DC = 2 \text{ cm}$, find the area of $\triangle BCD$.
- (iv) A ship sails from a port on a bearing of $235^\circ T$ for 150 nautical miles. It then turns and sails on a bearing of $120^\circ T$ until it reaches its destination due south of its original position. Calculate the distance of the ship from the port correct to three significant figures.

Question 5: (Start a new Booklet)

- (i) Find the size of $\angle EDC$,
giving precise reasons,
given $AB \parallel CD$.

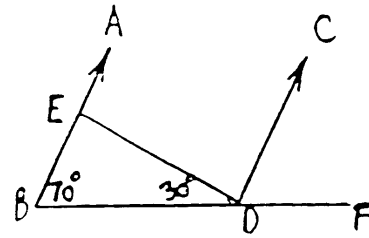


Diagram not to scale

- (ii) Given, $ST \parallel QR$

$$ST = 9 \text{ cm}$$

$$QR = 15 \text{ cm}$$

$$TR = 3 \text{ cm}$$

- (a) Copy this diagram into your Examination Booklet.

- (b) Prove: $\triangle PST \parallel \triangle PQR$.
give precise reasons

- (c) Hence, find the length of PT.
give precise reasons

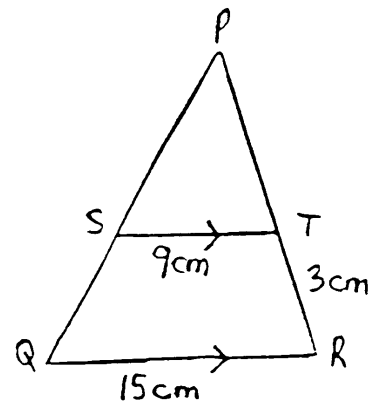


Diagram not to scale

- (iii) ABCD is a parallelogram such that

$$AP = PQ = QC$$

Copy this diagram into your Examination Booklet.

Prove that $DP = BQ$.
give precise reasons

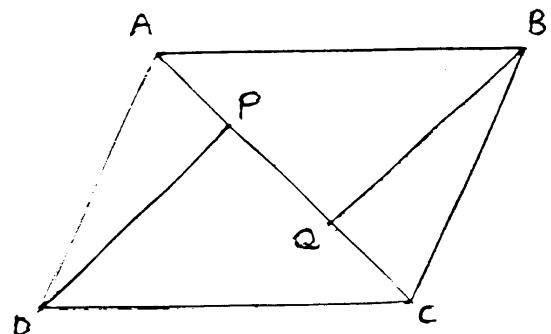


Diagram not to scale

Question 6: (Start a new Booklet)

- (i) Sketch: $y = e^{-x}$, stating its domain and range.
- (ii) Find the area between $y = e^{-x}$, the x-axis and the lines $x = -1$ and $x = 1$.
Give the answer correct to three significant figures
- (iii) Solve, $x^4 - 3x^2 + 2 = 0$.
- (iv) From a set of 40 discs numbered 1 to 40, one disc is selected at random.
What is the probability that the number is a multiple of either 5 or 7?
- (v) In a certain arithmetic progression, the fourth term is 7, and the fourteenth term is 32.
- (a) Find the first term and the common difference.
- (b) Calculate the sum of the first 50 terms.

Question 7: (Start a new Booklet)

- (i) Evaluate: $\sum_{n=1}^{\infty} 4 \times \left(\frac{1}{3}\right)^{n-1}$
- (ii) For $y = 2x^3 - 24x$
- (a) determine the stationary points
- (b) determine the nature of these stationary points
- (c) determine points of inflexion
- (d) sketch showing all important features.

Question 8: (Start a new booklet)

- (i) At a seaside holiday resort, a fisherman is selling fish. He claims that all the fish are fresh, but in actual fact, only 70% of the fish are fresh.
- (a) A woman chooses **TWO** fish at random. What is the probability that both fish are fresh?
- (b) A tourist chooses **THREE** fish at random. What is the probability that at least one of the fish is fresh?
- (ii) Find the equation of the tangent at the point $\log_e 2$ on the curve $y = \log_e(x+1)$.
- (iii) For the parabola, $x^2 + 32y = 0$, find:
- (a) the focal length
- (b) the co-ordinates of the focus
- (iv) Derive the equation of the locus $P(x,y)$ which is equidistant from $A(3,1)$ and $B(2,5)$.

Question 9: (Start a new Booklet)

- (i) Sketch: $y \leq 3 - x^2$, showing all main features.
- (ii) Evaluate: $\int_1^4 (x^2 + 1) dx$.
- (iii) Use Simpson's Rule with 2 equal subintervals to find an approximation for the area between the curve $y = x^2 + 1$, the x-axis and $x = 1$ and $x = 4$.
- (iv) ABCDE is a pentagon of fixed perimeter P cm. Its shape is such that ABE is an equilateral triangle and BCDE is a rectangle. If the length of AB is x cm:
- (a) Show that the length BC is $\frac{P - 3x}{2}$ cm
- (b) Show that the area of the pentagon is given by:
- $$A = \frac{1}{4}(2Px - (6 - \sqrt{3})x^2)$$
- (c) Find the value of $\frac{P}{x}$ for which the area of the pentagon is a maximum.

Question 10: (Start a new Booklet)

- (i) Sketch the area between the curve $y = \sqrt{x}$, the y-axis and $y = 1$ and $y = 2$.
- (ii) Find the volume of the solid obtained by rotating the area in (i) about the y-axis.
- (iii) Find an expression for the volume of the solid obtained by rotating the area in (i) about the x-axis. **DO NOT EVALUATE.**

- (iv) (a) Differentiate: e^{x^2} .
- (b) Hence, or otherwise, find: $\int x e^{x^2} dx$.

- (v) A loan of \$6000 is to be repaid by equal annual instalments.
Compound Interest, calculated yearly is 9% p.a. If the annual instalment is \$P, then:

$$A_1 = \$ (6000 \times 1.09 - P) \quad \text{is the amount owing at the end of ONE year.}$$

$$A_2 = \$ (6000 \times (1.09)^2 - P(1 + 1.09)) \quad \text{is the amount owing at the end of TWO years.}$$

- (a) Write an expression for A_n , the amount owing at the end of n years.
- (b) If the \$6000 loan (including interest charges) is exactly repaid at the end of n years, write an expression for the annual instalment, \$P.
- (c) Calculate the annual instalment, \$P, when $n = 25$.

END OF PAPER

VI. LIMIT EXAM. SOLUTIONS

Q1

(i) $I = \frac{3(18)}{1.5+2.5}$
 $= 13.5$

(ii) $m_{AB} = \frac{1+3}{3-2} = -4/5$

b) $m_{AC} = -4/5$
 $\theta = 141^\circ$

(iii) $\frac{x^2+x-12}{x^2-9} = \frac{(x-3)(x+4)}{(x-3)(x+3)}$

$= \frac{x+4}{x+3}$

(iv) $\frac{1}{3-5} + \frac{1}{3+12}$

$= \frac{3+12}{4 \cdot 15} + \frac{3-5}{4 \cdot 2} = \frac{6}{7}$

(v) $|3x-7|=8$

$3x-7=8 \quad 3x-7=-8$

$3x=15 \quad 3x=-1$

$x=5 \quad x=-1/3$

(vi) $x = 0.011$

$\log x = 4.141$

$\log x = 4.1$

$x = \frac{416}{440}$

Q2

(i) $3x^2 + 2x - 3 = 0$

a) $x+p = -4/3$

b) $x+p = -3/3 = -1$

c) $x^2+p^2 = (-4/3)^2 - 2(-1)$
 $= 16/9 + 2$
 $= 37/9 \quad (34/9)$

(ii) a) $\frac{\sin 60}{\cos 60} = \frac{\sqrt{3}}{1/2} = \sqrt{3}$

b) $\cos 225^\circ = -1/\sqrt{2}$

c) $\lim_{x \rightarrow 27} \frac{x^2-27}{x-27} = 4$

$\lim_{x \rightarrow 27} \frac{(x-3)(x^2+3x+9)}{x-27} = 37$

(i) $d = \frac{14(-1) + 3(2)}{\sqrt{1+13}}$

$= \frac{-14+6}{5} = \frac{-8}{5} = -4$

$\therefore$ d of line from origin -4

Q3

(i) $4 \tan \theta = -3$

$\tan \theta = -3/4$

$\theta = 143.3^\circ, 323.3^\circ$

(ii) $x^2 - (12+4)x + (7+6) = 0$

for equal roots $\Delta = 0$

$(12+4)^2 - 4(7+6) = 0$

$k^2 + 8k + 16 - 28 - 4k = 0$

$k^2 + 4k - 12 = 0$

$(k+6)(k-2) = 0$

$k = 2 \text{ or } -6$

(iii) a) $\int \frac{1}{x^3} dx = -\frac{1}{2x^2} + C$

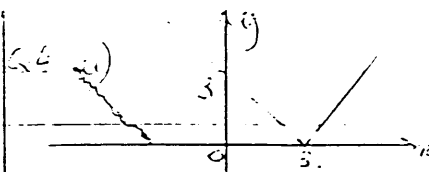
b) $\int (5x-2) dx = \frac{(5x-2)^2}{20}$

(iv) a) $y' = \frac{(1+3x) \cdot 1 - (2+5) \cdot 3}{(1+3x)^2}$

$= \frac{1+3x-3x-15}{(1+3x)^2}$

$= \frac{-14}{(1+3x)^2}$

b) $y' = \frac{1}{4-3x} = -3$
 $= \frac{-3}{4-3x}$



(i) $\frac{4-7}{3} > 2 - \frac{4}{5}$

$45-54 > 30-37$

$15 > 27$

$4 < 7\frac{1}{2}$

(ii) a) $\cos C = \frac{6^2+6^2-8^2}{2 \cdot 6 \cdot 6}$

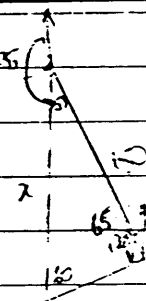
$C = 83.57^\circ$

b) $\angle BAC = 45.2^\circ$

$\Delta = A_{\triangle ABC} = \frac{1}{2} \cdot 28 \sin 45.2^\circ$
 $= 5.46 \text{ cm}^2$

(iii) $x = \frac{150}{\sin 65^\circ}$

$x = \frac{150 \sin 65^\circ}{\sin 60^\circ}$
 $= 156.477$



$\angle BAC = 110^\circ$ (const. $\angle$)
 $\therefore \angle BDC = 110.30^\circ$ (adj. $\angle$)
 $= 80^\circ$

(iv) $\angle P$ common

$\angle PQR = \angle PRQ$ (corr. $\angle$ - alt)

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$\therefore \triangle PQR \parallel \triangle PRQ$ (AA)

$\frac{PT}{9} = \frac{PT+3}{15}$

$15PT = 9PT + 27$

$6PT = 27$

$PT = 4\frac{1}{2}$

iii) $\angle OAF = \angle AOB$ (alt. $\angle$'s)

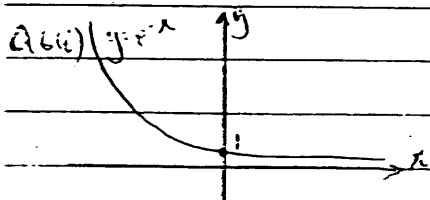
$AO = BO$ (radii of circle)

$AP = OQ$ (given)

$\therefore \triangle APO \cong \triangle BOQ$ (SAS)

$\therefore \angle POA = \angle QOB$ (corresponding angles)

of comp. $\angle$'s)



D: $x \in \mathbb{R}$ R: $y > 0$

$$\begin{aligned} \text{ii) } A &= \int_{-5}^0 e^{-x} dx \\ &= [-e^{-x}]_{-5}^0 \\ &= (-e^{-0} + e^{-5}) \\ &= e^{-5} - 1 \end{aligned}$$

$$\text{iii) } x^4 - 3x^2 + 2 = 0$$

$$(x^2 - 2)(x^2 - 1) = 0$$

$$x = \pm \sqrt{2}, \pm 1$$

$$\text{iv) } P(\text{sum is } 5, 7) = \frac{12}{40} = \frac{3}{10}$$

$$\text{v) } T_4 = a + 3d = 7$$

$$T_{14} = a + 13d = 32$$

$$\therefore 10d = 25$$

$$d = 2\frac{1}{2}$$

$$\therefore a = 7 - 3(2\frac{1}{2})$$

$$= -\frac{1}{2}$$

$$\text{a) } x = -\frac{1}{2} \quad a = 2\frac{1}{2}$$

$$\text{b) } S_{50} = \frac{50}{2} [-1 + 49(2\frac{1}{2})] = 3037\frac{1}{2}$$

Q7

$$\text{a) } \sum_{n=1}^{\infty} 4 \times (\frac{1}{3})^{n-1}$$

$$a = 4 \quad r = \frac{1}{3}$$

$$S_{\infty} = \frac{a}{1-r} = \frac{4}{1-\frac{1}{3}} = \frac{4}{\frac{2}{3}} = 4 \times \frac{3}{2} = 6$$

$$\text{ii) } y = 3x^2 - 24x$$

$$\text{c) } y' = 6x - 24$$

$$\therefore 6x^2 - 24x = 0 \quad \text{for local pt.}$$

$$x^2 - 4x = 0$$

$$x = 0, 4$$

$$\text{b) } y' = 12x \Rightarrow x = 2 \quad y' = 24 \Rightarrow \text{min}$$

$$x = 2 \Rightarrow y = -32 \quad \text{min at } (2, -32)$$

$$x = 2 \Rightarrow y = 32 \quad \text{max at } (-2, 32)$$

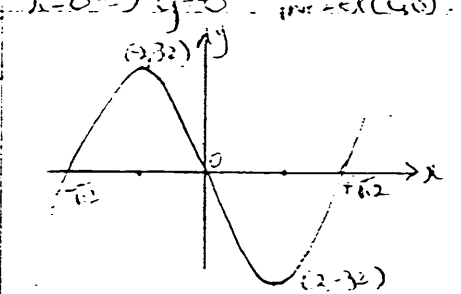
$$\text{c) } y'' = 0 \Rightarrow 12x = 0$$

$$\therefore x = 0$$

$$\text{check } -1 \quad y' = -12 \quad \text{decreasing}$$

$$1 \quad y' = 12 \quad \text{increasing}$$

$$x = 0 \Rightarrow y = 0 \quad \text{inflection at } (0, 0)$$



$$2x^3 - 24x = 0 \Rightarrow 2x(x^2 - 12) = 0$$

$$x = 0, \pm \sqrt{12}$$

$$\text{Q8. (i) a) } P(\text{fresh}) = 0.7 \times 0.7$$

$$= 0.49$$

$$\text{b) } P(\text{not fresh}) = 1 - P(\text{fresh})$$

$$= 1 - (0.7)^3$$

$$= 0.473$$

$$\text{a) } y = \ln(x+1)$$

$$y' = \frac{1}{x+1}$$

$$\text{at } (1, \ln 2) \quad m = \frac{1}{1+1} = \frac{1}{2}$$

$$y - \ln 2 = \frac{1}{2}(x - 1)$$

$$x - 2y + \ln 2 - 1 = 0$$

$$\text{ii) } x^2 + 32y = 0 \Rightarrow x^2 = -32y$$

$$\therefore 4u = -32$$

$$u = -8$$

$$\text{c) } \text{focal length} = -8$$

$$\text{d) } \text{focus } (0, -8) \quad \text{axis at } (0, 0)$$

$$\text{iv) } D_m = \sqrt{(x-3)^2 + (y-1)^2}$$

$$D_{PB} = \sqrt{(x-2)^2 + (y-5)^2}$$

$$D_m = D_{PB} \Rightarrow D_m^2 = D_{PB}^2$$

$$\text{i.e. } x^2 - 6x + 9 + y^2 - 2y + 1 =$$

$$x^2 - 4x + 4 + y^2 - 10y + 25$$

$$2x - 8y + 19 = 0$$

$$2x - 8y + 19 = 0$$

$$\text{Q9 a) } \text{Graph of } y = 3 - x^2$$

$$\text{b) } \int_{-3}^3 (x^2 + 1) dx = [\frac{x^3}{3} + x]_{-3}^3$$

$$= \frac{6^3}{3} + 4 - \frac{-3^3}{3} - 1$$

$$= 24$$

$$\text{c) } A = \frac{2\frac{1}{2}}{3} [2 + 17 + 4(2\frac{1}{2} + 1)]$$

$$= 40$$

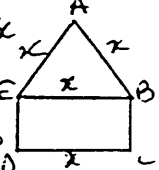
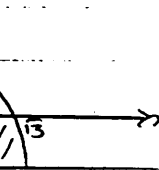
$$\text{d) } P = x + x + x + 28x$$

$$\therefore Bx = \frac{P - 3x}{2}$$

$$\text{b) } A = Bx \cdot x + \frac{1}{2} x \cdot x \sin 60$$

$$= \frac{P - 3x}{2} \cdot x + \frac{x^2}{2} \cdot \frac{\sqrt{3}}{2}$$

$$= \frac{2Px - 6x^2 + \sqrt{3}x^2}{4} = \frac{1}{4} (2Px - (6 - \sqrt{3})x^2)$$



$$\frac{dH}{dx} = \frac{1}{4} (2P - 2(6-\sqrt{3})x)$$

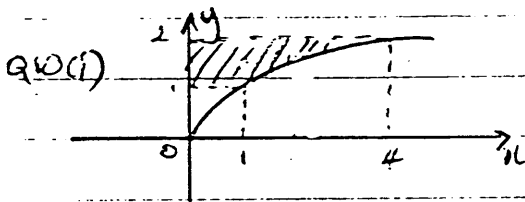
$$\text{So } 2P - 2(6-\sqrt{3})x = 0$$

$$P - (6-\sqrt{3})x = 0$$

$$P/x = 6-\sqrt{3}$$

$$\frac{d^2H}{dx^2} = \frac{-2(6-\sqrt{3})}{4} < 0$$

concave up
 So $P/x = 6-\sqrt{3}$ is a ^{value} ~~max~~
 for area to be max



$$\text{ii) } V = \pi \int_1^2 y^4 dy$$

$$= \pi \left[\frac{y^5}{5} \right]_1^2$$

$$= \pi \left(\frac{32}{5} - \frac{1}{5} \right)$$

$$= \frac{31\pi}{5} u^3$$

$$\text{iii) } V = 4\pi - \pi + 4\pi \cdot 3 - \pi \int_1^4 x dx$$

$$= 3\pi + 12\pi - \pi \int_1^4 x dx$$

$$= 15\pi - \pi \int_1^4 x dx$$

$$\text{(iv) a) } \frac{d}{dx} e^{x^2} = 2x e^{x^2}$$

$$\text{(b) } \int 2x e^{x^2} dx = \frac{1}{2} \int 2x e^{x^2} dx$$

$$= \frac{1}{2} e^{x^2} + C$$

$$\text{(v) a) } A_n = \$ (6000(1.09)^n - \text{P})$$

$$P(1 + 1.09 + \dots + (1.09)^{n-1})$$

$$\text{b) } P = \frac{6000(1.09)^n}{1 + 1.09 + \dots + (1.09)^{n-1}}$$

$$\text{c) } P = \frac{6000(1.09)^{25}}{\frac{(1.09)^{25} - 1}{1.09 - 1}}$$

$$= \$610.84$$